Unit 2
Virtualization

Virtualization is a technology that uses software to create virtual versions of physical computing resources, such as servers, storage, networks, and operating systems.
It allows multiple virtual machines (VMs) to run on a single physical machine, enhancing hardware utilization and enabling efficient resource allocation.
Key benefits include cost reduction, improved scalability, increased energy efficiency, and better disaster recovery, making it a foundational technology for cloud computing.

➢ VIRTUALIZATION REFERENCE MODEL

Virtualization Reference Model (VRM) is a conceptual framework that desribes how virtuliaztion components interact in a cloud computing environment.

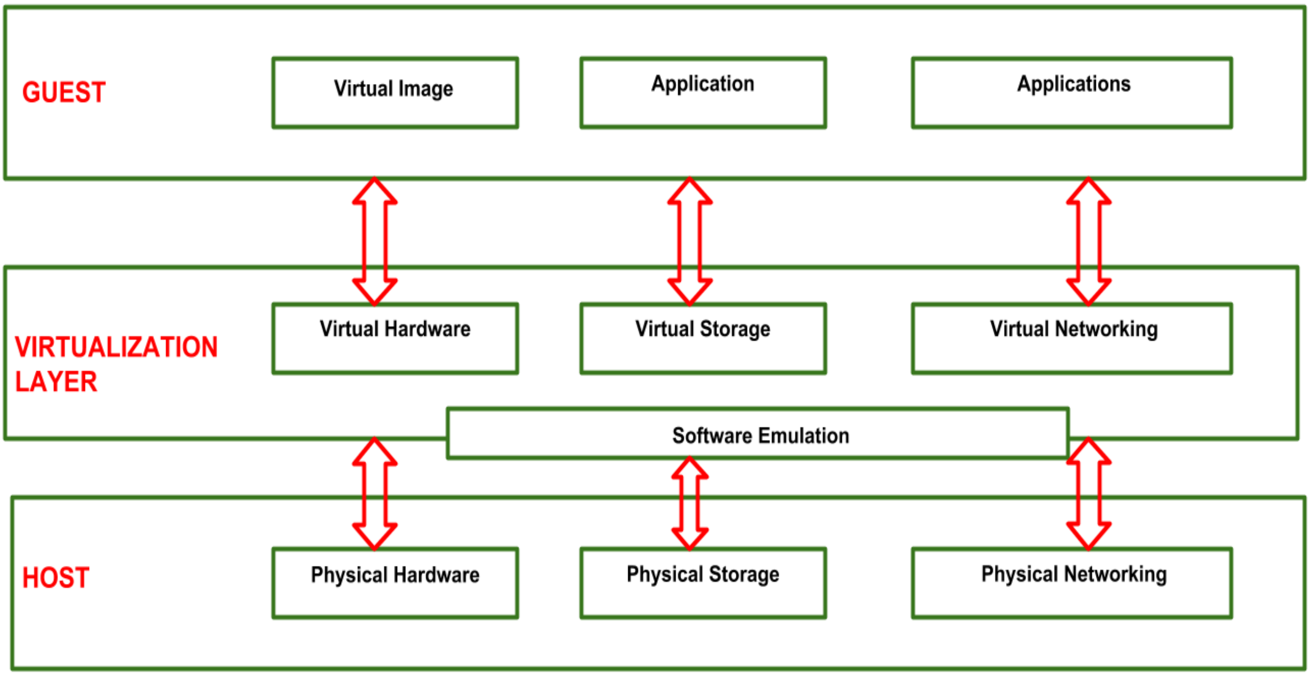


Figure: Reference Model of Virtualization.

There are the main layers of VRM : -

1. HOST
This is the physical foundation, consists of the **actual hardware** like **processors**, **memory**, **storage**, and **network devices**.
The **operating system** on the host is often responsible for managing these physical resources.

2. Virtualization Layer (Hypervisor):
Hypervisor is also called as **Virual Machine Monitor**.
This is the **core** of the model, a **software** layer that **creates** and **manages virtual machines**.
It **abstracts** the **physical resources**, presenting a **virtualized version** to the guests.
The hypervisor intercepts requests from the guests and translates them for the physical hardware.
The virtualization layer provides the guest with a virtualized version of the hardware, giving the guest the impression that it has exclusive access to a complete system.

Types of hypervisors:

Type 1 Hypervisor (Bare-Metal Hypervisor):

The hypervisor is **installed directly** on the underlying host system.
It does **not require** any **base server operating system**.
It has direct access to **hardware resources**.
Examples :
VMware ESXi, Citrix XenServer, and Microsoft Hyper-V hypervisor.

Type 2 Hypervisor:
It is run over an **installed operating system** (such as Windows or macOS).
Examples :
VMware Workstation, Oracle VirtualBox.

3. GUEST
The guest represents the system **component** that **interacts** with the **virtualization layer** rather than with the **host**.
The guest is the software component, most often an operating system, that runs within the virtual environment.
Instead of interacting directly with the physical hardware (the "host"), the guest interacts with the virtualization layer.
The main purpose of the guest is to execute applications and perform tasks within its isolated virtual environment, allowing multiple guests to share the same physical hardware.

Characteristics Of Virtualization

Hardware Abstraction & Resource Pooling:
Virtualization abstracts the underlying physical hardware, allowing IT managers to create a virtual layer over physical resources like servers, storage, and networks. These pooled resources can then be dynamically allocated to virtual machines (VMs) or virtual instances as needed.

Sharing:
Multiple virtual machines or computing environments can share the same host physical machine and its resources (CPU, memory, storage). This improves hardware utilization and reduces the number of physical servers required, leading to cost and energy savings.

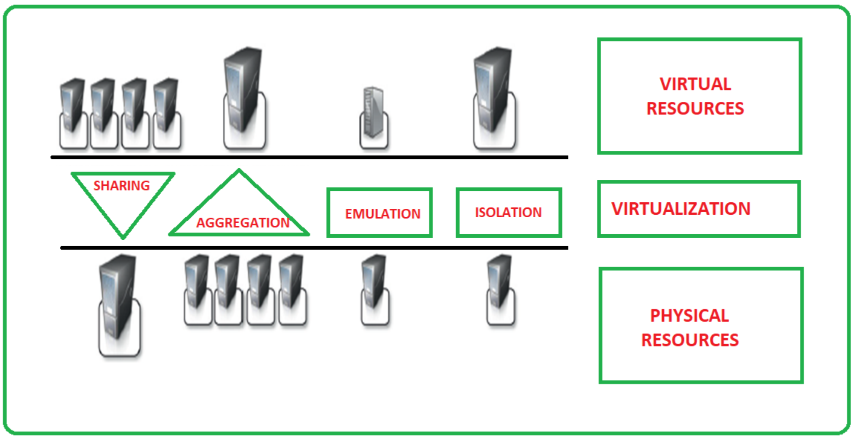
Isolation of Resources
Each virtual machine is isolated from the others, and they cannot access each other's resources. This provides security and stability, as a problem with one virtual machine will not affect the others.

Emulation:
Emulation means using a program or a device to imitate the working of another program or device. A program/device completely different to the host can be emulated on the host device through virtualization.

Flexibility and Scalability:
Virtual machines can be easily created, modified, and destroyed, providing significant flexibility to adapt to changing demands. Resources can be scaled up or down quickly without needing new physical hardware, a concept known as elasticity.

Portability:
Virtualization also provides portability, as virtual machines can be easily moved between physical machines. This allows for easy disaster recovery, as virtual machines can be quickly moved to a different physical machine in the event of a disaster. It also allows for easy migration between physical machines, as virtual machines can be moved to new hardware without affecting the applications or data.

Aggregation:
In contrast to sharing, aggregation combines multiple separate physical hosts into a single, unified virtual resource that appears as one large server to the user. This allows for greater consolidation of resources.



Types of Virtualization

Storage Virtualization:

Storage virtualization refers to the process of combining storage from multiple physical storage devices (like hard drives, SSDs, and storage arrays) into a single, unified, logical storage resource. Such integration is done to enhance efficient working from the system, better security, and easier management.

It ensures smooth performance and efficient operations even when the underlying hardware changes or fails.

Example:

Amazon S3 is an example of storage virtualization because in S3 we can easily store any amount of data from anywhere.

Suppose a MNC have lots of files and data of company to store. By Amazon S3 company can store all their files and data in one place and access these from anywhere without any kind of issue in secure way.